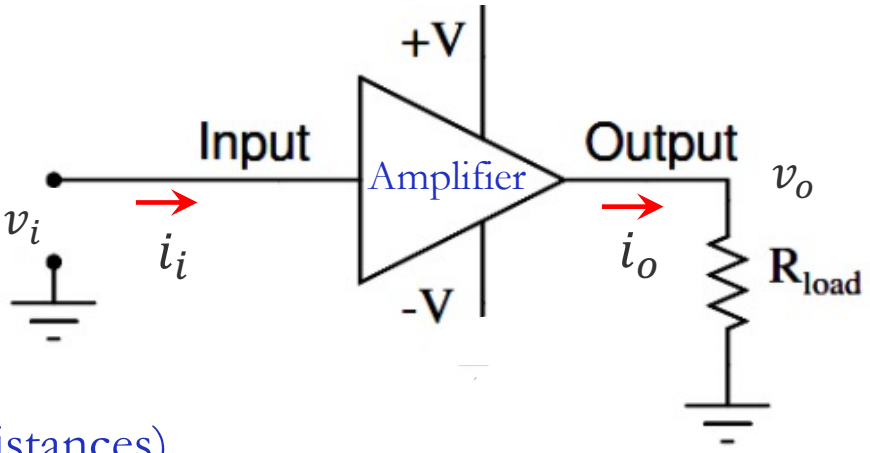
Field Effect Transistors

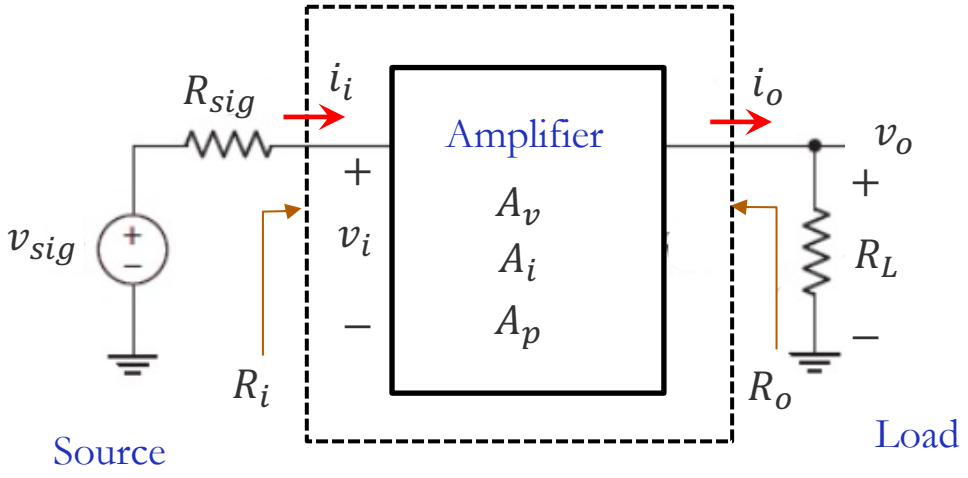
- The MOSFET as an Amplifier.
- Small Signal Analysis.

Amplifiers: Introduction

Basic requirements of any amplifier are:

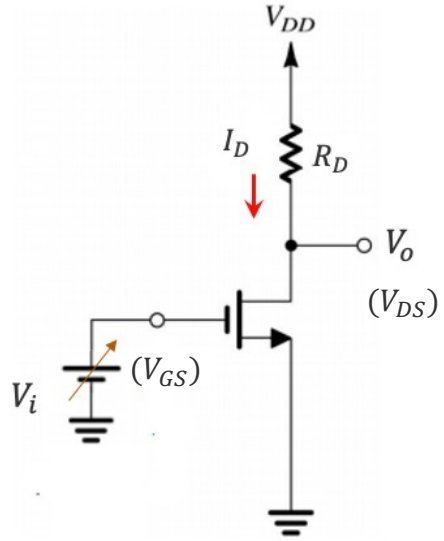
1) Gain (voltage, current, power)
2) Linear operation.
3) Large swing (dynamic range).
4) Suitable coupling (input and output resistances).

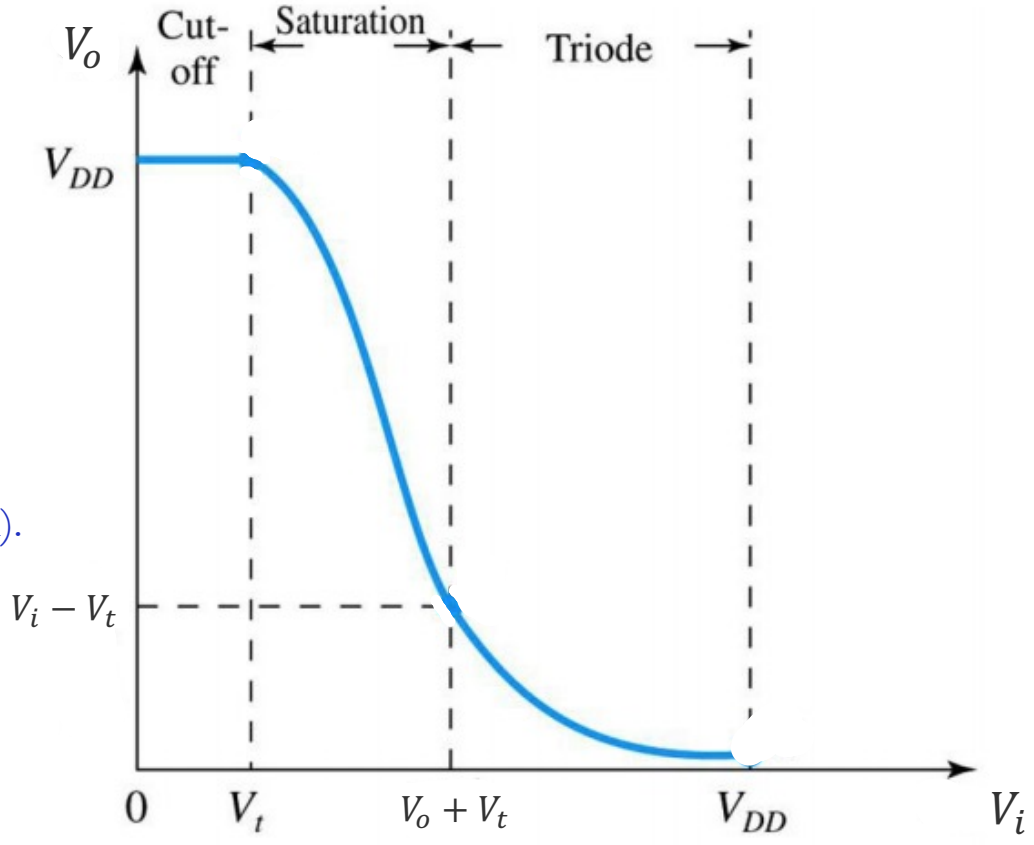




MOSFET Transfer Function

Consider the following DC transistor circuit.





1) V_i (small) $< V_t$: Cutoff $I_D = 0, V_o = V_{DD}$ (high).

2) $V_i > V_t : I_D \neq 0, \quad V_o = V_{DD} - I_D R_D$

V_o is still high but decreasing

Saturation

3) V_i (high) $\gg V_t$: $I_D \neq 0, \quad V_o = V_{DD} - I_D R_D$

V_o has decreased too much

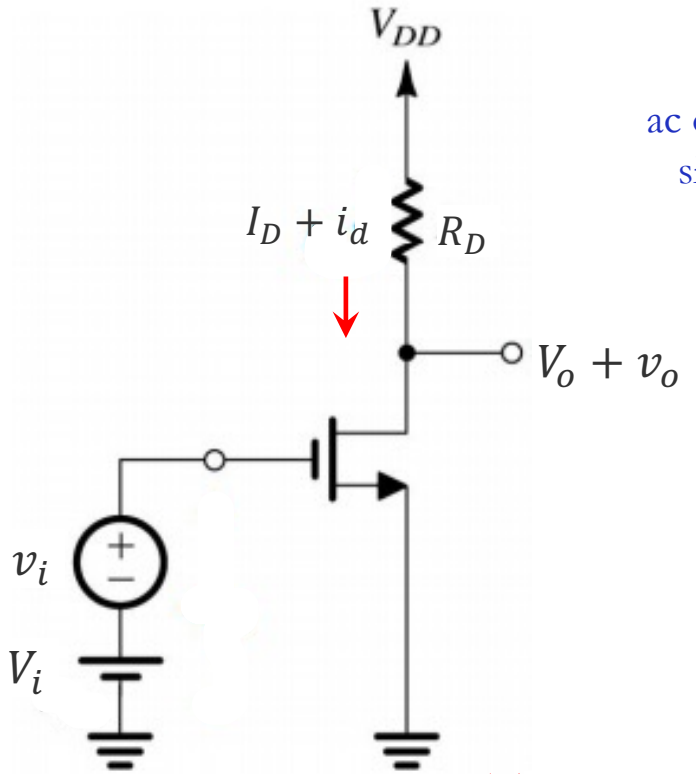
Triode

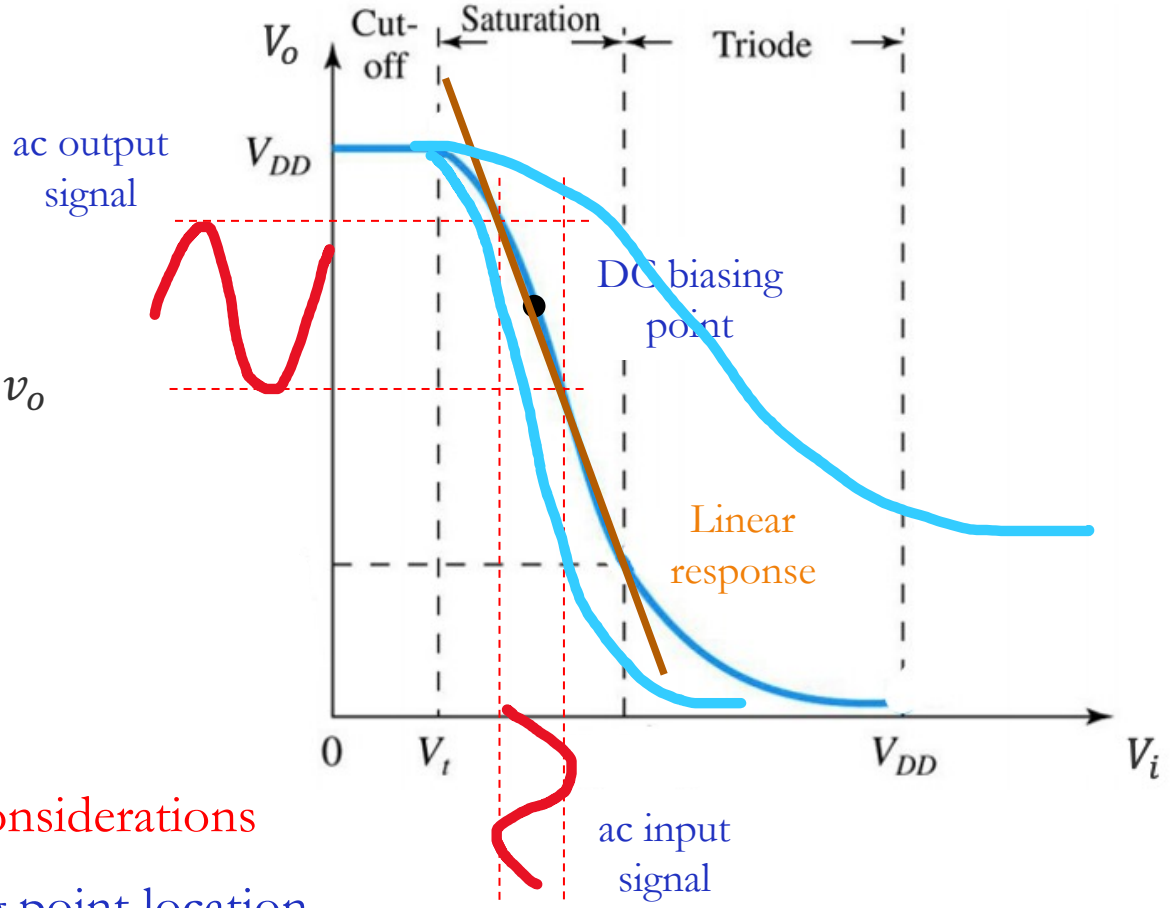
Which mode is suitable for amplification?

Remember the basic requirements.

MOSFET Transfer Function

Now an ac source is added at the input.





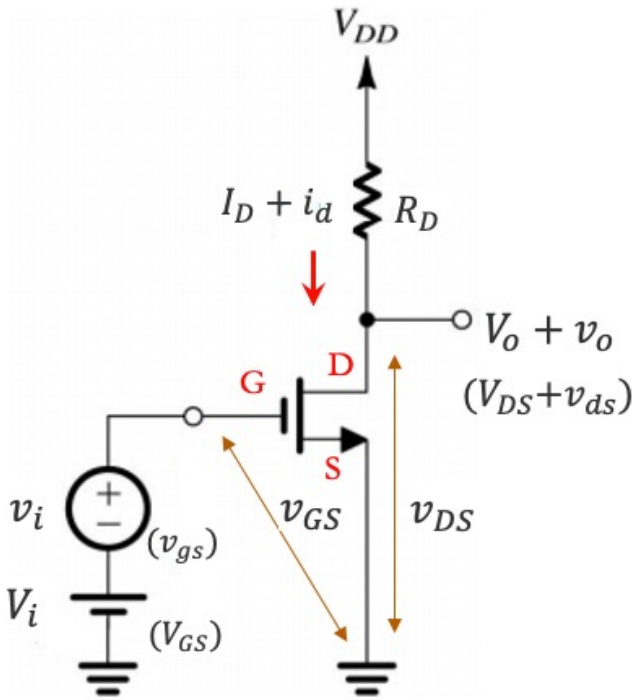
Two considerations

1) Biasing point location

2) How large should the ac signal be?

The MOSFET as an Amplifiers

Reconsider the previous circuit. Let's write the current equation.



Note the symbol convention
DC: all capital
ac: all small
Total: mixed

The transistor is in saturation.

$$i_D = k(v_{GS} - V_t)^2 \quad \Rightarrow \quad i_D = k\left(V_{GS} + v_{gs} - V_t\right)^2$$

$$i_D = \underbrace{k(V_{GS} - V_t)^2}_{I_D \text{ (DC part)}} + \underbrace{2k(V_{GS} - V_t)v_{gs} + k v_{gs}^2}_{i_d \text{ (ac part)}}$$

For a small input signal (small-signal approximation)

$$i_D = I_D + i_d = k(V_{GS} - V_t)^2 + 2k(V_{GS} - V_t)v_{gs}$$

$$i_d = 2k(V_{GS} - V_t)\, v_{gs} = g_m\, v_{gs}$$ i_d is linear with ac input

$$v_O = V_o + v_o = V_{DD} - (I_D + i_d)R_D = \underbrace{(V_{DD} - I_D R_D)}_{\text{(DC part)}} \underbrace{- i_d R_D}_{\text{(ac part)}}$$

$$v_o = -i_d\, R_D = -g_m R_D\, v_{gs}$$ v_o is linear with ac input

$$i_d = g_m\, v_i$$

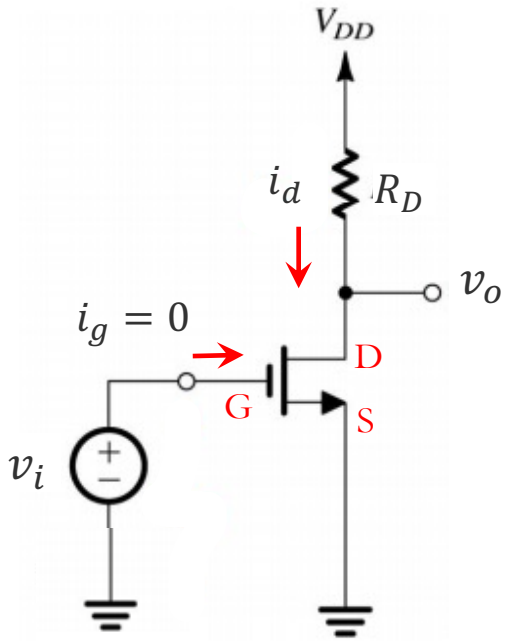
$$v_o = -g_m R_D\, v_i$$

$$g_m = 2k(V_{GS} - V_t) = 2\sqrt{k}\sqrt{I_D}$$

Transconductance (A/V)

The MOSFET as an Amplifiers

Important conclusions regarding the operation of the MOSFET as an amplifier.



1) The amplifier voltage gain is given as v_o/v_i.

For this circuit $\dfrac{v_o}{v_i} = -g_m R_D$

2) The gain is a function of the DC status (bias).

3) The transconductance g_m is a function of the DC status (bias) and transistor dimensions.

4) The transistor's small-signal current and voltage relations can be converted to a "circuit model".

$i_d = g_m\, v_i$

$v_o = -g_m R_D\, v_i$

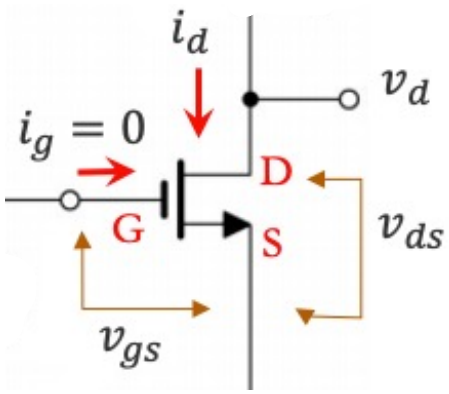
$g_m = 2k(V_{GS} - V_t) = 2\sqrt{k}\sqrt{I_D}$

$i_g = 0$ ➡ Open circuit

$i_d = g_m\, v_{gs}$ ➡ Controlled current source

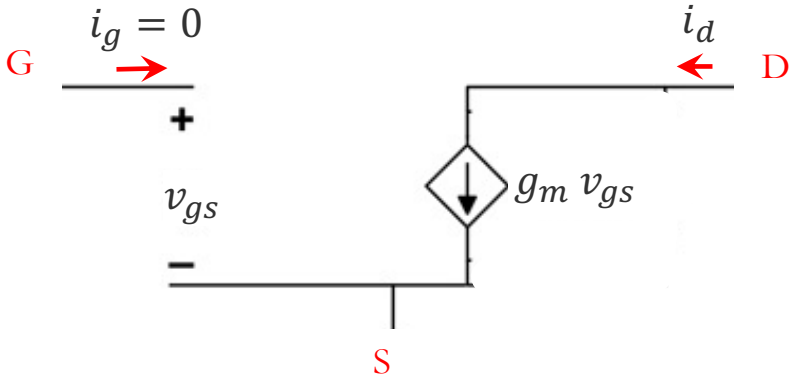
$\lambda \neq 0$ ➡ Output resistance

MOSFET Small-signal Circuit Model

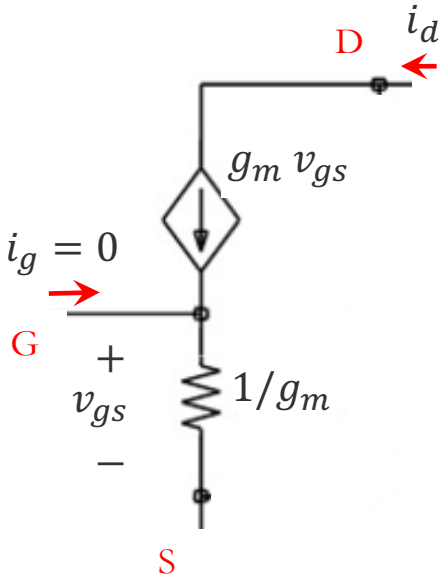


The transistor's small-signal current and voltage relations can be converted to a "circuit model".

$i_g = 0$	➡	Open circuit
$i_d = g_m\, v_i$	➡	Controlled current source
$\lambda \neq 0$	➡	Output resistance $r_o = V_A / I_D$



The NMOS π Model



The NMOS T Model

Small-signal Analysis

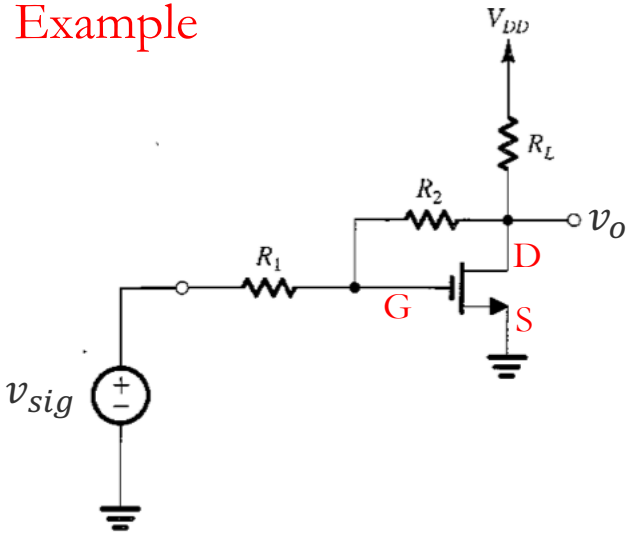
Procedure

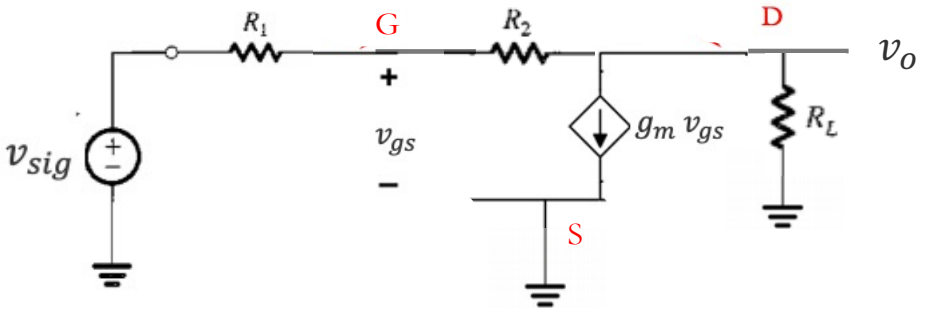
1) Perform DC analysis of the circuit. Find I_D and hence g_m and r_o.

2) Produce the small-signal equivalent circuit using the pi or T model of the transistor.

3) Analyze the circuit using circuit equations (KVL, KCL, etc.) to find important amplifier parameters (gain, input/output resistances, etc.)

Example





Small-signal equivalent circuit

Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.